

SARDAR PATEL UNIVERSITY**M. Sc. Information Technology****Semester – I External Examinations****PS01CINT03 – Introduction to Theoretical Computer Science****Saturday, 15th April, 2017****Time: 10:00 a.m. to 01:00 p.m.****Total Marks: 70****Q-1 Give most appropriate answer of following multiple choice questions:****[8]**

1. A set $X = \{1, 3\}$ is a _____ set.
(A) finite (B) infinite
(C) uncountable infinite (D) none of these
2. A relation which is reflexive, symmetric and transitive is known as _____ relation.
(A) unbounded (B) equivalence
(C) triple (D) none of these
3. R be a relation defined on a set A, then R is _____ if $b R a$ holds whenever $a R b$ holds for $b \in A$. In other words if $(a, b) \in R \Rightarrow (b, a) \in R$.
(A) reflexive relation (B) symmetric relation
(C) asymmetric relation (D) relative relation
4. The degree of a vertex having one self loop is _____.
(A) 1 (B) 2
(C) 3 (D) 4
5. A path having same initial and terminal vertices is also forms _____.
(A) open walk (B) elementary walk
(C) circuit (D) None of these
6. A lattice is _____ if meet and join operation distributes over each other.
(A) reflexive (B) parallel
(C) distributive (D) none Of these
7. Increase in price of milk during world war is known as _____ variation.
(A) irregular (B) regular
(C) cyclic (D) secular
8. If the value of some variable lies in range, rather than a fix number, then it is an example of _____ variable.
(A) crisp (B) fuzzy
(C) static (D) none of these

Q-2 Answer the following questions (Any Seven):**[14]**

1. What are countable infinite sets? Give one example.
2. Define rule of sum and give its example.
3. Define poset and give its example.
4. Explain isomorphic graph with example.
5. Define algorithm and state characteristics of good algorithm.
6. Define intractable problem and give its example.
7. For any a and b in A , prove that $a \vee (a \wedge b) = a$.
8. List all the components of the time series.
9. Differentiate between fuzzy set and crisp set.

Q-3

[A] If $f(x)=8x+3$ and $g(x)=7x-5$ then find $f \circ g(x)$, $g \circ f(x)$, $f \circ g(3)$ and $g \circ f(4)$.

[6]

[B] Define and give one example for each of the following:

[6]

(i) reflexive relation (ii) transitive relation and (iii) anti-symmetric relation.

OR

[B] Explain with example hamiltonian path and hamiltonian circuit.

[6]**Q-4**

[A] Discuss shortest path algorithm in detail.

[6]

[B] Write a detail note on complexity of problems with example.

[6]**OR**

[B] For any a, b, c, d in a lattice $(A, \leq)$ if $a \leq b, c \leq d$ then prove that

[6]

(i) $a \vee c \leq b \vee d$ and (ii) $a \wedge c \leq b \wedge d$.

Q-5

[A] Explain boolean algebra and boolean lattice in detail with example.

[6]

[B] Determine the trend values for the following data using moving average method for 3 years.

[6]

Year	1980	1981	1982	1983	1984	1985	1986	1987	1988	1989	1990
Product	5	10	15	20	25	30	35	40	45	50	55

OR

[B] Define time series and discuss utilities of time series in detail.

[6]**Q-6**

[A] List the methods for measurements of time series component? Explain any one of them.

[6]

[B] Write a detailed note on fuzzy logic and its applications.

[6]**OR**

[B] Let fuzzy set $P = \{(x_1, 0.7)(x_2, 0.2)(x_3, 0)\}$ and $Q = \{(x_1, 0.4)(x_2, 0.8)(x_3, 1)\}$ be two fuzzy sets defined on universe of discourse X . Perform the following operations

(i) $P \cup Q$ (ii) $P \cap Q$ (iii) Q' .

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